

openQCM Next Impedance Analysis: measuring shifts in frequency and bandwidth based on calculation of conductance G

Background and theory

The measurement of the quartz impedance is crucial in QCM sensing application because it is directly related to the shifts in frequency and bandwidth (or dissipation) [1]. The impedance analysis is based on the conductance curve G analysis: the series resonance frequency f_r corresponds to the peak of the conductance curve and Γ is the “half bandwidth at half height” (“bandwidth” for simplicity) of the conductance curve. According to Johannsmann, Γ is used here as the quantity to account for dissipative processes.

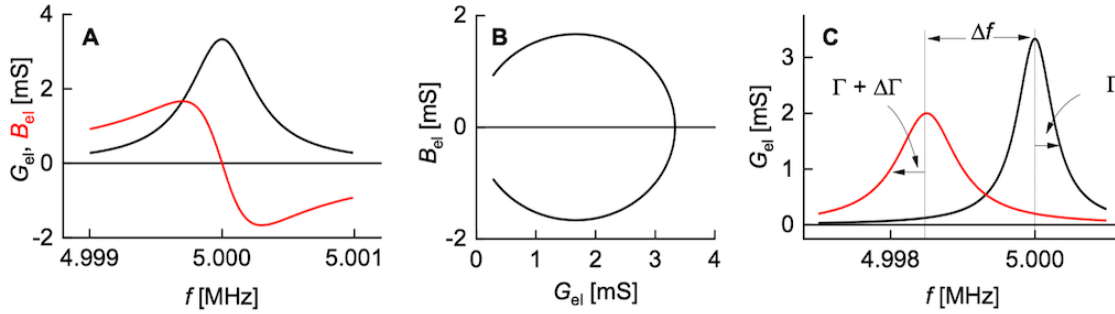


Figure A. Typical output impedance analysis, showing: A) conductance (black) and susceptance (red), B) the polar diagram and C) Shifts in frequency and bandwidth, as described by Johannsmann in [1].

Kanazawa – Gordon (KG) theory predicts a shift in resonance and bandwidth when one liquid, i.e. a semi – infinite viscoelastic medium, replaces another on the face of a quartz crystal sensor [REF]. The shift in frequency Δf and bandwidth $\Delta\Gamma$ is given by:

$$\Delta f_n = -f_0^{3/2} \sqrt{n \frac{\eta_L \rho_L}{\pi \eta_q \rho_q}} \quad (A1)$$

$$\Delta\Gamma_n = f_0^{3/2} \sqrt{n \frac{\eta_L \rho_L}{\pi \eta_q \rho_q}} \quad (A2)$$

Where: f_0 is the fundamental resonance frequency of the “bare” quartz sensor, η_L is the liquid viscosity, ρ_L is the liquid density, η_q is the quartz shear modulus, ρ_q is the quartz density and $n = 1, 3, 5, \dots$ denotes the odd overtone index. Looking at the equation A1 and A2 it is important to underline some qualitative characteristics. When one face of a quartz is in contact with a liquid medium the KG theory predicts that:

1. The shift in resonance frequency and bandwidth is equal and opposite, no matter what the liquid and the overtone are
2. The shift in resonance frequency and bandwidth scale as $\sqrt{n}$
3. The shift in resonance frequency and bandwidth is proportional to the square root of the product liquid density - viscosity $\sqrt{\eta_L \rho_L}$

The impedance circuit and calculation of conductance

The impedance analysis of openQCM Next is based on the voltage divider circuit formed by the QCM sensor and the resistor R17 connected to ground, and the usage of the component AD8302 for measuring gain loss and phase across the QCM sensor, **Figure 1**.

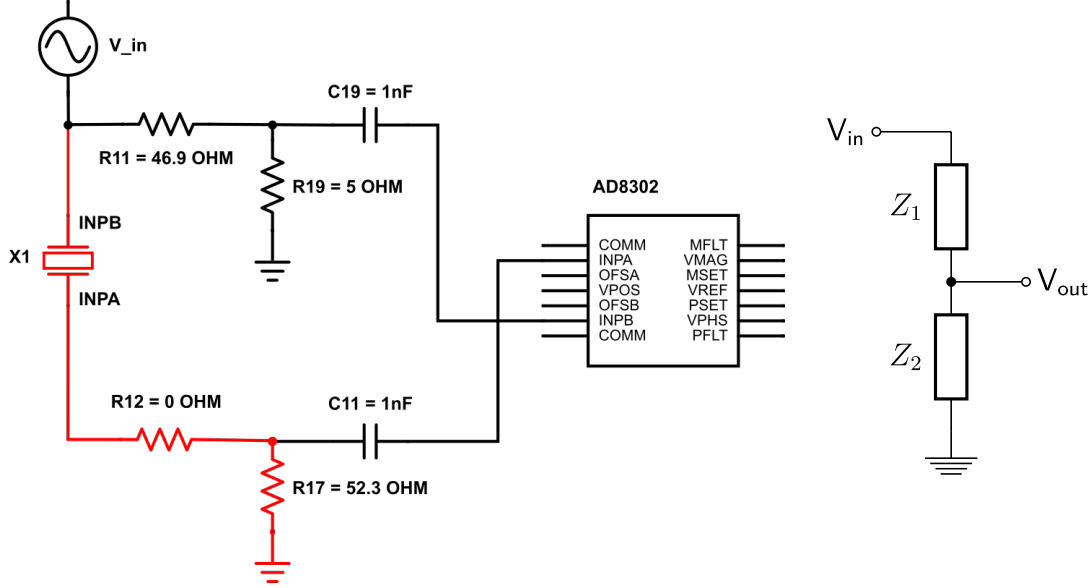


Figure 1. Left: openQCM Next voltage divider circuit (in red) and the component AD8302 measuring gain loss and phase. Right: A simple voltage divider circuit, general case.

In general case a voltage divider referenced to ground is created by connecting two electrical impedances in series. The transfer function, known as voltage ratio, is given by:

$$H = \frac{V_{out}}{V_{in}} = \frac{Z_2}{(Z_1 + Z_2)} \quad (1)$$

Coming back to the openQCM Next voltage divider circuit, the quartz impedance modulus $|Z_q|$ is given by:

$$|Z_q| = R_{17} \left(\frac{V_{INPB}}{V_{INPA}} - 1 \right) \quad (2)$$

The AD8302 is an IC component capable of measuring gain loss and phase [2]. The AD8302 output consist of DC voltages, named V_{MAG} and V_{PHS} , related to gain loss and phase according to the transfer function:

$$V_{MAG} = R_F I_{SLP} \log \left(\frac{V_{INA}}{V_{INB}} \right) + V_{CP} \quad (3a)$$

$$V_{PHS} = -R_F I_\phi (|\Phi(V_{INA}) - \Phi(V_{INB})| - 90^\circ) + V_{CP} \quad (3b)$$

where: the slope of the gain function $R_F I_{SLP}$ is 600 mV/decade, the slope of the phase function $-R_F I_\phi$ is 10 mV/degree, and the center point V_{CP} is 900 mV. The idealized transfer characteristic for the gain and phase is represented in **Figure 2**

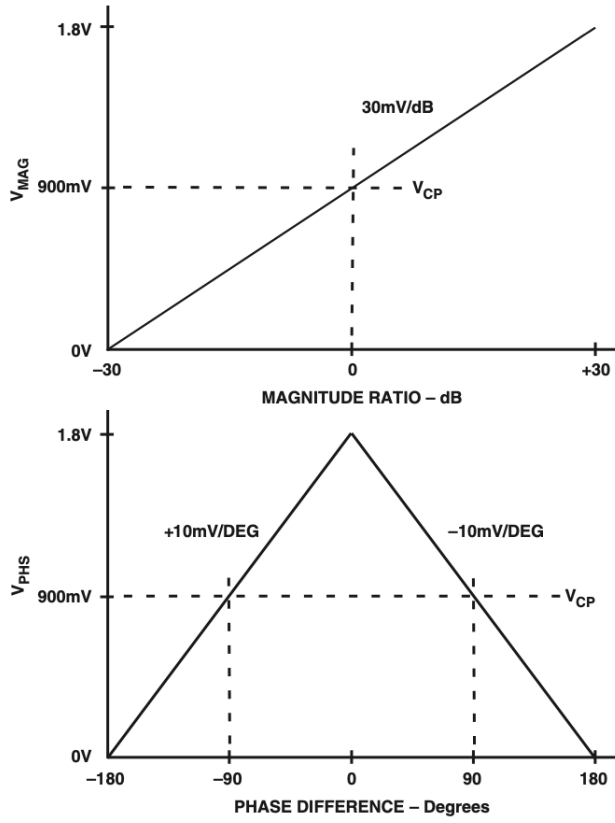


Figure 2. AD8302 idealized transfer characteristic for gain loss and phase. With a center point of 900 mV in 0 dB gain loss, a range from -30 dB to 30 dB covers the V_{MAG} output from 0 V to 1.8 V. With a center point of 900 mV in 90° , a range of 0° to 180° covers the the V_{PHS} output from 0 V to 1.8 V

Doing some math on equation (3a), the voltage ratio is given by:

$$\frac{V_{INB}}{V_{INA}} = 10^{\left(\frac{V_{CP} - V_{MAG}}{R_{FISLP}}\right)} \quad (4)$$

Back the openQCM Next voltage divider circuit, combining equation (2) and (4), it is possible to calculate the **quartz impedance modulus** $|Z_q|$ as:

$$|Z_q| = R_{17} \left(10^{\left(\frac{V_{CP} - V_{MAG}}{R_{FISLP}}\right)} - 1 \right) \quad (5)$$

and from equation (3a), it is possible to write the **quartz impedance phase** ϕ as:

$$\phi = |\phi_{INA} - \phi_{INB}| = \frac{V_{CP} - V_{PHS}}{R_F I_\phi} + 90^\circ \quad (6)$$

Impedance $\vec{Z}$ and its inverse the admittance $\vec{Y}$ are complex number defined as:

$$\vec{Z} = R + jX \quad (7a)$$

$$\vec{Y} = G + jB = \frac{1}{\vec{Z}} \quad (7b)$$

One can calculate the **conductance** G , the real part of admittance, as:

$$G = \frac{R}{R^2 + X^2} \Rightarrow \quad (8)$$

$$G(V_{MAG}, V_{PHS}) = \frac{|Z_q| \cos \phi}{|Z_q|^2} = \frac{\cos \phi (V_{PHS})}{|Z_q| (V_{MAG})}$$

In conclusion, by measuring the output voltage of AD8302 V_{MAG} and V_{PHS} it is possible to elaborate the conductance curve $G(V_{MAG}, V_{PHS})$ based on the equation (8). The analysis of the conductance curve allows the measure of shifts in QCM frequency and bandwidth, as discussed before.

Monitoring shifts in frequency and dissipation on different overtones caused by the passage air – water

The impedance analysis is tested using the **openQCM Next equipment**, monitoring the response of a 5 MHz 14 mm quartz sensor to the passage air to water, at a constant temperature of 25°C. The results compared with the theoretical values of Kanazawa Gordon equation, generalized on higher overtones equation (A1) e (A2)

The output voltage of AD8302 V_{MAG} and V_{PHS} were collected by sweeping around the resonance frequencies at each overtone, with a frequency sweep range of 18 KHz and frequency step of 1 Hz. The collected raw data were filtered, using a Savitzky-Golay - spline processing, then equation (8) was applied to calculate the conductance spectrum up to the 9-th overtone Figures 3A and 3B.

From a qualitative point of view, it is evident the drastic variation of the conductance spectrum for the passage from air to water, showing a typical lowering - broadening of the resonance curves caused by the drastic variation in the viscoelastic properties.

Figure 3B shows how the resonance frequency is detected, as the peak of the conductance curve, and how the bandwidth is detected, as the half bandwidth at half height, showed for the fundamental resonance.

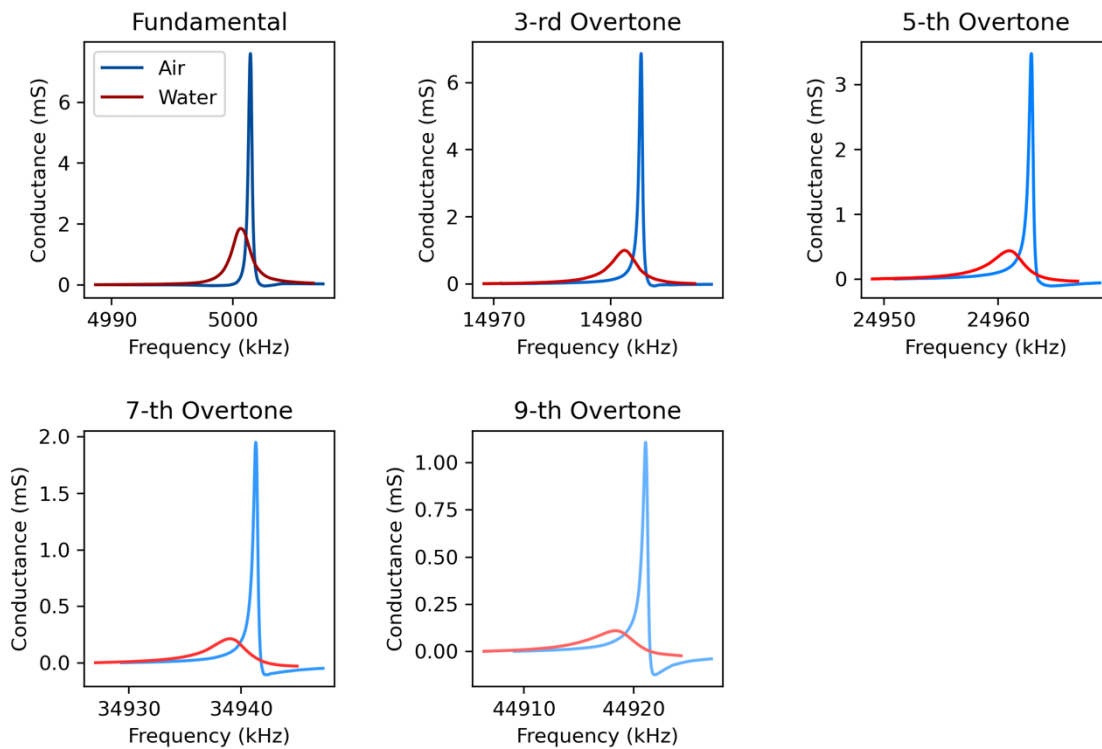


Figure 3A Conductance spectrum of a 5 MHz 14 mm quartz sensor up to the 9-th overtone, for the passage from air to water.

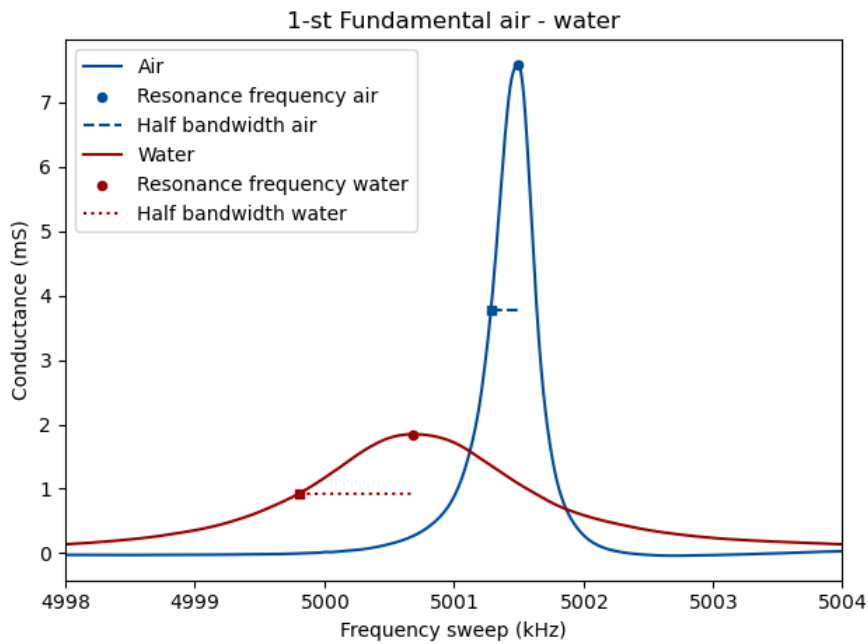


Figure 3B Processing the conductance curve of a 5 MHz 14 mm quartz sensor at the fundamental resonance for measuring the shifts in frequency and half bandwidth caused by the passage from air to water.

The openQCM Next device was used for real – time monitoring of shift in frequency and bandwidth of a 5 MHz 14 mm AU-coated QCM sensor, temperature controlled at 25°C, caused by the passage from air to water up to the 9th overtone, Figure 4. Normalized shifts in frequency $\Delta f_n/n$ and bandwidth $\Delta \Gamma_n/n$ have been compared with Kanazawa Gordon equation (A1) and (A2), using water density 1.102933 g/cm³ and viscosity 0.010665 g/(cm·s) , Figure 5 [REF TO WATER RHO ETA]

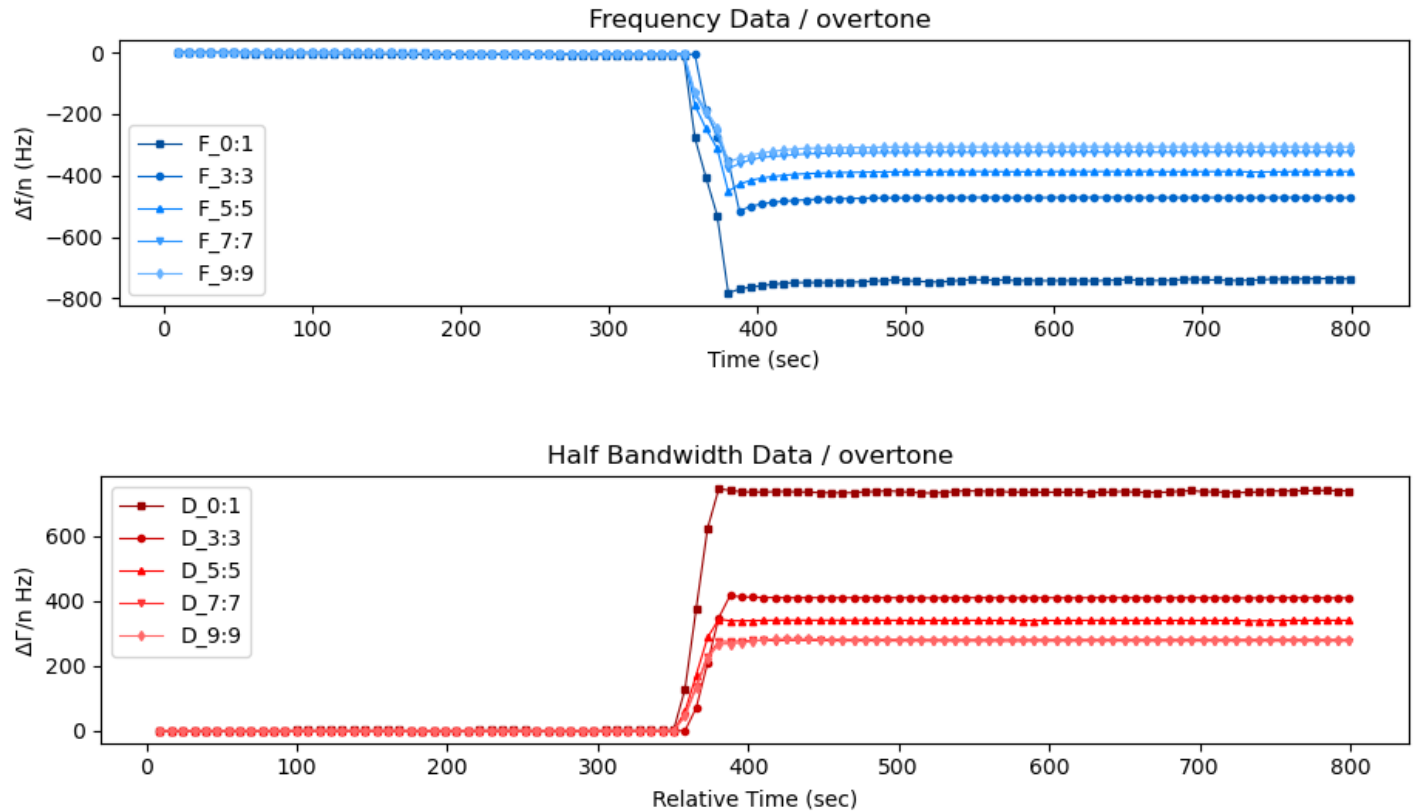


Figure 4 Acquired data of normalized frequency (blue) and half – bandwidth (red) shifts caused by the passage from air to water

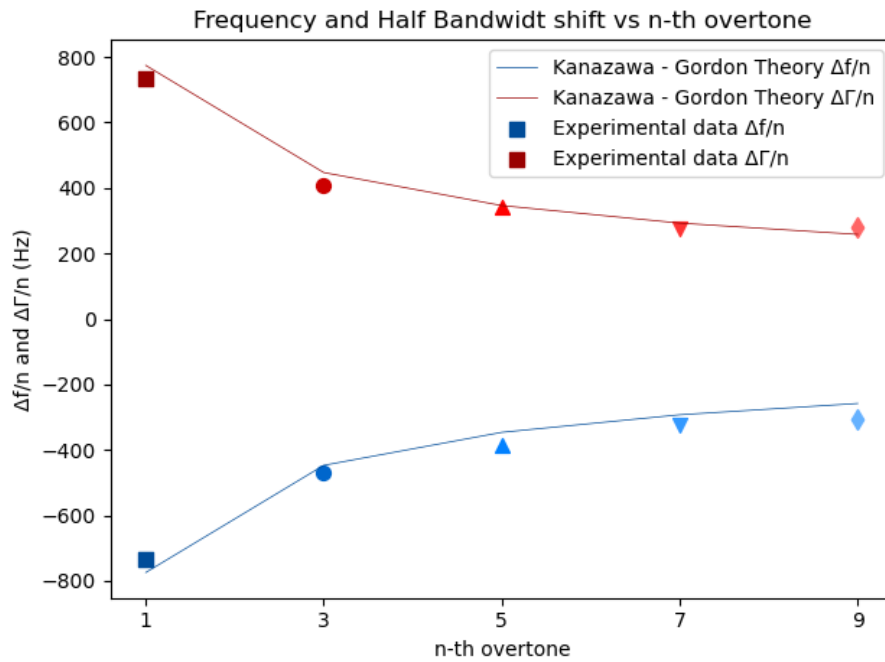


Figure 5 Normalized frequency (blue) and half – bandwidth (red) shifts experimental data compared to the prediction of Kanazawa – Gordon equation.

Overtone n	Frequency shift / n	Bandwidth shift / n	KG Theory
1	-736	734	770
3	-469	409	445
5	-385	340	344
7	-322	276	291
9	-304	281	256

Table 1. Data of normalized frequency (blue) and half – bandwidth (red) shifts experimental data compared to the prediction of Kanazawa – Gordon equation.

In conclusion, data acquired using the impedance analysis scheme responds quite well to the KG qualitative characteristics as:

1. The shift in resonance frequency and bandwidth are (quite) equal and opposite, regardless from the overtone
2. The shift in resonance frequency and bandwidth scale as $\sqrt{n}$

And most importantly the quantitative results are in good agreement with the prediction of KG equation.

References

- [1] *Sensors* **2021**, *21*(10), 3490; <https://doi.org/10.3390/s21103490> by Diethelm Johannsmann
- [2] AD8302 LF–2.7 GHz RF/IF Gain and Phase Detector <https://www.analog.com/en/products/ad8302.html>

Acknowledgements

Many thanks to Dr. Qifeng Wang, Research Associate at Northwestern University, an expert in polyelectrolytes, adhesion mechanics, and the quartz crystal microbalance. His valuable and generous contribution, in terms of knowledge, ideas, python coding just to name a few, has been fundamental to this development.